Assignment 5

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Please note that

- **Released date: March 19th, Tuesday.**
- **Due date: March 28th, Thursday, by 11:59pm.**
- Late submission is **NOT** accepted.
- Please submit your answers as a PDF file with a name like "120010XXX ASS5.pdf" (Your student ID + ASS No.). You may either typeset you answers directly using computers, or scan your handwritten answers. (We recommend you use the printers on campus to scan. If you use your smartphone to scan, please limit the file size 10MB.)

Question 1. Consider the matrix

$$A = \begin{bmatrix} 1 & 2 & 3 & 0 & 1 & 2 \\ 3 & 2 & 1 & 0 & -1 & -2 \\ 0 & -3 & -2 & 0 & 3 & 2 \\ 1 & 3 & 2 & 3 & 1 & 1 \end{bmatrix}$$

1. Calculate the rank and nullity of A.
2. Determine basis of

(a) $\mathrm{Row}(A)$, the row space of A;

(b) $\mathrm{Col}(A)$, the column space of A;

(c) $\mathrm{Null}(A)$, the null space of A.

Solution

1. Transforming A to RREF, we find

$$B = \begin{bmatrix} 1 & 0 & 0 & 0 & 1/2 & 0 \\ 0 & 1 & 0 & 0 & -2 & -2 \\ 0 & 0 & 1 & 0 & 3/2 & 2 \\ 0 & 0 & 0 & 1 & 7/6 & 1 \end{bmatrix}$$

Then

$$\begin{aligned}\dim \mathrm{Row}(A) &= \#\text{ non-zero rows of } B\\ &= 4\end{aligned}$$

Thus

$$\mathrm{rank}(A) = 4$$

By Rank-Nullity Theorem,

$$\begin{aligned}\mathrm{rank}(A) + \ &\text{nullity }(A) = 6\\ &\text{nullity }(A) = 6 - 4 = 2.\end{aligned}$$

2.

(a) The non-zero rows of B form a basis of Row (A); hence

$$\begin{aligned}\Big\{&(1,0,0,0,1/2,0)^\top, (0,1,0,0,-2,-2)^\top,\\ &(0,0,1,0,3/2,2)^\top, (0,0,0,1,7/6,1)^\top\Big\}\end{aligned}$$

is a basis of Row (A).

(b) The columns of A corresponding to the pivot columns of B form a basis of Col(A) :

$$\left\{ a_1 = \begin{bmatrix}1\\3\\0\\1\end{bmatrix}, a_2 = \begin{bmatrix}2\\2\\-3\\3\end{bmatrix}, a_3 = \begin{bmatrix}3\\1\\-2\\2\end{bmatrix}, a_4 = \begin{bmatrix}0\\0\\0\\3\end{bmatrix} \right\}$$

(c)

$$\begin{aligned}\text{Null }(A) &= \big\{x \in \mathbb{R}^6 \mid Bx = 0\big\}\\ &= \text{Span}\left(\begin{bmatrix}-1/2\\2\\-3/2\\-7/6\\1\\0\end{bmatrix}, \begin{bmatrix}0\\2\\-2\\-1\\0\\1\end{bmatrix}\right)\end{aligned}$$

where we set two vectors as u and v, respectively.

Since $\dim N(A) = \text{nullity }(A) = 2$, we get $\{u, v\}$ is a basis of Null(A).

Question 2. Suppose that $A \in \mathbb{R}^{m\times n}$.

1. Show that, if Null$(A) = \{0\}$, then

(a) $\text{Row}(A) = \mathbb{R}^n$.

(b) $m \geq n$.

(c) A is full rank.

2. Show that, if $Ax = b$ is consistent for all $b \in \mathbb{R}^m$, then $n(A) = n - m$.

3. Show that, if $m \geq n$ and $\exists y \in \mathbb{R}^n, y \neq 0$, $Ay = 0$, then $\exists b \in \mathbb{R}^m, Ax = b$ is not consistent.

Solution

1. Suppose that $\text{Null}(A) = \{0\}$. Then, $n(A) = \dim \text{Null}(A) = 0$.
By, Rank-nullity theorem:

$$\begin{aligned} & \text{rank}(A) + n(A) = n \\ (*) \quad & \text{rank}(A) = n. \end{aligned}$$

(a) $\text{rank}(A) = \dim \text{Row}(A)$ and $\text{Row}(A)$ is a subspace of $\mathbb{R}^n$; thus

$$(*) \quad \Rightarrow \text{Row}(A) = \mathbb{R}^n.$$

[A basis B of $\text{Row}(A)$ must consist of n vectors; n linearly independent vectors in $\mathbb{R}^n$ form a basis of $\mathbb{R}^n$].
(b) $\text{rank}(A) = \dim \text{Col}(A)$ and $\text{Col}(A)$ is a subspace of $\mathbb{R}^m$. So $m \geq \dim \text{Col}(A) = n$.
(c) Since $m \geq n$, we see that

$$\text{rank}(A) = n = \min(m, n).$$

2. Suppose that

$$(*') \left\{ \ \forall b \in \mathbb{R}^m, \quad \text{the system } Ax = b \text{ is consistent.} \ \right\} \tag{1}$$

- Let $A = \begin{bmatrix} a_1 & a_2 & \cdots & a_n \end{bmatrix}$, where $a_1, a_2, \cdots, a_n$ are the column vectors of A.
- For $b \in \mathbb{R}^m$, recall that $Ax = b$ is consistent $\Leftrightarrow b \in \text{Span}\,(u_1, \ldots, u_n)$
- Thus,

$$(*') \Rightarrow \text{Span}\,(u_1, \cdots, u_n) = \mathbb{R}^m.$$

- We have $\text{Col}(A) = \text{Span}\,(u_1, \ldots, u_n) = \mathbb{R}^m$. Thus $\text{rank}(A) = \dim \text{Col}(A) = m$, and

$$\begin{aligned} & n(A) + \text{rank}(A) = n \\ & n(A) = n - \text{rank}(A) = n - m. \end{aligned}$$

3. Suppose that $m \geq n$ (so $n - m \leq 0$), and

$$\exists y \in \mathbb{R}^n, \quad y \neq 0, \quad Ay = 0.$$

Then

$$\odot \quad y \in \text{Null}(A) \quad \text{and} \quad y \neq 0. \tag{2}$$

- $\odot \Rightarrow \dim \text{Null}(A) \geq 1 \Rightarrow n(A) \geq 1$.
- By part 2,

$$n(A) \geq 1 \Rightarrow n(A) \neq n-m (\text{by } m \geq n) \Rightarrow \left\{ \exists b \in \mathbb{R}^n, \ Ax = b \text{ is inconsistent} \right\}$$

(Otherwise, if $\forall b \in \mathbb{R}^m, Ax = b$ is consistent, then $n(A) = n - m$ by part 2).

Question 3. Consider $A \in \mathbb{R}^{4\times 3}$ and consider $S = \{Ax \mid x \in \mathbb{R}^3\}$.

1. Verify that, if $y \in S$, then $y \in \mathbb{R}^4$.
2. Show that S is a subspace of $\mathbb{R}^4$.
3. Show that $S = \text{Col}(A)$.
4. Let $\{e_1, e_2, e_3\}$ be the standard basis of $\mathbb{R}^3$. Let $y_1 = Ae_1, y_2 = Ae_2$ and $y_3 = Ae_3$. Show that

$$S = \text{Span}\,(y_1, y_2, y_3)\,.$$

5. Suppose that $\text{rank}(A) = 3$.

(a) Show that $\text{Null}(A) = \{0\}$.

(b) Show that $\{y_1, y_2, y_3\}$ is a basis of the column space of A.

(c) Let B be the reduced row echelon form of A. Determine B.

Solution

1. Suppose that $y \in S$. Then

$$\exists x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \in \mathbb{R}^3, \quad y = Ax$$

where $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{bmatrix} = \begin{bmatrix} a_1 & a_2 & a_3 \end{bmatrix}$.

Then, Ax is 4×1 matrix $\in \mathbb{R}^4$ because A is a 4×3 matrix and x is a 3×1 matrix.

2.

- Let $x = 0 \in \mathbb{R}^3$. Then $y = Ax \in S$. But $y = Ax = A0 = 0$, so $0 \in S$.
- Let $y_1, y_2 \in S$, and let $z = y_1 + y_2$. Then, $\exists x_1, x_2 \in \mathbb{R}^3, y_1 = Ax_1, y_2 = Ax_2$. Let $x = x_1 + x_2$. Then

$$Ax = Ax_1 + Ax_2 = y_1 + y_2 = z$$

and $Ax \in S$. Thus $z \in S$.
- Let $y \in S$ and $\alpha \in \mathbb{R}$. Then

$$\exists x \in \mathbb{R}^3, \quad y = Ax.$$

Thus

$$\alpha y = \alpha Ax = A(\alpha x) \in S$$

because $\alpha x \in \mathbb{R}^3$.

3. Denote by u_1, u_2 and u_3 the column vectors of A. Then $\text{Col}(A) = \text{Span}\,(u_1, u_2, u_3)$.
- Consider $y \in \mathbb{R}^4$. Notice that,

$y \in S \Leftrightarrow \left\{ \text{ The system } Ax = y \text{ is consistent } \right\}$.

[In fact, if $y \in S$, then (*) $\exists x \in \mathbb{R}^3, \quad y = Ax$
so '$Ax = y$' is consistent;
and if '$Ax = y$' is consistent, then (*) holds which means that $y \in S$].
- By a theorem, '$Ax = y$' is consistent $\Leftrightarrow y \in \text{Span}\,(u_1, u_2, u_3)$.

4. We have

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, e_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, e_3 = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

$$A = \begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix}$$

$$u_1 = \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \\ a_{41} \end{bmatrix}, \quad u_2 = \begin{bmatrix} a_{12} \\ a_{22} \\ a_{32} \\ a_{42} \end{bmatrix}, \quad u_3 = \begin{bmatrix} a_{13} \\ a_{23} \\ a_{33} \\ a_{43} \end{bmatrix}$$

$$\bullet\ y_1 = Ae_1 = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \\ a_{41} & a_{42} & a_{43} \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} a_{11} \\ a_{21} \\ a_{31} \\ a_{41} \end{bmatrix} = u_1$$

$$y_2 = Ae_2 = u_2 \quad \text{and } y_3 = Ae_3 = u_3.$$

$$\bullet\ \mathrm{Col}(A) = \mathrm{Span}\,(u_1, u_2, u_3) = \mathrm{Span}\,(y_1, y_2, y_3)$$

and, by 3. $\mathrm{Col}(A) = S$.

5. Suppose that $\mathrm{rank}(A) = 3$.

(a) Rank-Nullity Theorem says

$$n(A) + \mathrm{rank}(A) = \#\text{ of columns of } A$$
$$= 3$$

$$n(A) = 3 - \mathrm{rank}(A) = 3 - 3 = 0$$

$$\dim \mathrm{Null}(A) = n(A) = 0$$

$$\mathrm{Null}(A) = \{0\}.$$

(b) $\mathrm{Col}(A) = S = \mathrm{Span}\,(y_1, y_2, y_3)$ and $\dim \mathrm{Col}(A) = \mathrm{rank}(A) = 3$, so $\{y_1, y_2, y_3\}$ is a basis of $\mathrm{Col}(A)$.

(c) let $B = \mathrm{RREF}(A)$.

• A is 4×3 matrix $\Rightarrow$ B is 4×3 matrix.

• $3 = \text{rank}(A) = \#$ non-zero rows of B.

So B has 0 zero row.

• $3 = \dim \text{Col}(A) = \#$ pivot column of B

so B has pivot columns. Thus

$$B = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix}$$

Question 4. Let $A \in \mathbb{R}^{m \times n}$. Show that

1. $\text{Null}(A) = \text{Null}\left(A^\top A\right)$
2. $\text{rank}(A) = \text{rank}\left(A^\top A\right)$.

Solution

1.

• Let $x \in \text{Null}(A)$. Then $x \in \mathbb{R}^n$ and

$$Ax = 0$$

Multiply by A^T :

$$A^\top Ax = 0$$

so $x \in \text{Null}\left(A^\top A\right)$.

• let $x \in \text{Null}\left(A^\top A\right)$. Recall that

$A \in \mathbb{R}^{m \times n}$ is $m \times n$ matrix

$A^\top \in \mathbb{R}^{n \times m}$ is $n \times m$ matrix

$A^\top A \in \mathbb{R}^{n \times n}$ is $n \times n$ matrix.

Thus, knowing that $x \in \text{Null}\left(A^\top A\right)$, we have

$$x \in \mathbb{R}^n \text{ and } A^\top Ax = 0.$$

Let $y = Ax$. Then, $y \in \mathbb{R}^m$ and

$$(*) \quad \begin{aligned} y^\top &= (Ax)^\top \\ &= x^\top A^\top \end{aligned}$$

Moreover, multiply,

$$A^\top Ax = 0$$

by $x^\top$; we get

$$x^\top A^\top Ax = 0$$

by (*) we know $x^\top A^\top = y^\top$ and we have $Ax = y$.

Then,

(**) $\quad y^\top y = 0$

Let $y = \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix}$. Then

$$0 = y^\top y = \begin{bmatrix} y_1 & \cdots & y_m \end{bmatrix} \begin{bmatrix} y_1 \\ \vdots \\ y_m \end{bmatrix} = y_1^2 + \cdots + y_m^2$$

which yields $y_1 = \cdots = y_m = 0$; i.e., $y = 0$.

Having proved that $y = 0$, we deduce that

$$Ax = 0$$

i.e. $x \in \text{Null}(A)$.

2. By rank-nullity theorem

• (on $A \in \mathbb{R}^{m \times n}$)

$$\text{rank}(A) + n(A) = n$$

• (on $A^\top A \in \mathbb{R}^{n \times n}$)

$$\text{rank}\left(A^\top A\right) + n\left(A^\top A\right) = n$$

Then,

$$\text{rank}(A) + n(A) = \text{rank}\left(A^\top A\right) + n\left(A^\top A\right)$$

where $n(A) = n\left(A^\top A\right)$ by part 2 ; hence $\text{rank}(A) = \text{rank}\left(A^\top A\right)$.

Question 5. Let $\quad A = \begin{bmatrix} 1 & -5 & 2 \\ 1 & -2 & 1 \\ 2 & 4 & -1 \end{bmatrix}$.

1. Calculate the following minors of A :

$$M_{21}, M_{22} \text{ and } M_{23}$$

2. Calculate the following cofactors of A :

$$A_{21}, A_{22} \text{ and } A_{23}$$

3. Calculate $\det(A)$.

Solution

1. M_{ij} is obtained by deleting the ith row and j^{th} column of A.

- Thus

$$M_{21} = \begin{bmatrix} -5 & 2 \\ 4 & -1 \end{bmatrix}, \quad M_{22} = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}, M_{23} = \begin{bmatrix} 1 & -5 \\ 2 & 4 \end{bmatrix}$$

2. $A_{ij} = (-1)^{i+j} \det(M_{ij})$.

- Thus

$$A_{21} = (-1)^{2+1} \begin{vmatrix} -5 & 2 \\ 4 & -1 \end{vmatrix} = -(5-6) = 1$$

$$A_{22} = (-1)^{2+2} \begin{vmatrix} 1 & 2 \\ 2 & -1 \end{vmatrix} = -1-4 = -5$$

$$A_{23} = (-1)^{2+3} \begin{vmatrix} 1 & -5 \\ 2 & 4 \end{vmatrix} = -(4+10) = -14$$

3. Expanding along the 2nd row:

$$\begin{aligned} \det(A) &= a_{21}A_{21} + a_{22}A_{22} + a_{23}A_{23} \\ &= 1 \times 1 + (-2) \times (-5) + 1 \times (-14) \\ &= 1 + 10 - 14 = -3 \end{aligned}$$

Question 6. Calculate the determinant of the matrix

$$A = \begin{bmatrix} 4 & 3 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 7 & 4 & 1 & 1 & 0 \\ 5 & 2 & 0 & 0 & 1 \\ 0 & -1 & 0 & 1 & -1 \end{bmatrix}$$

Solution

- Expand along 3 column :

$$\begin{aligned} \det(A) &= 0 + 0 + (-1)^{3+3} \begin{vmatrix} 4 & 3 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 5 & 2 & 0 & 1 \\ 0 & -1 & 1 & -1 \end{vmatrix} \\ &= \det(B) \end{aligned}$$

- Expand along 2 row of B :

$$\det(B) = 0 + 0 + 0 + (-1)^{2+4} \begin{vmatrix} 4 & 3 & 1 \\ 5 & 2 & 0 \\ 0 & -1 & 1 \end{vmatrix} = \det C$$

- Expand along 3rd row of C :

$$\begin{aligned} \det(C) &= 0 + (-1)(-1)^{2+3} \begin{vmatrix} 4 & 1 \\ 5 & 0 \end{vmatrix} \\ &+ (1)(-1)^{3+3} \begin{vmatrix} 4 & 3 \\ 5 & 2 \end{vmatrix} \\ &= -5 - (8 - 15) \\ &= 10 - 8 = 2 \end{aligned}$$

Question 7. Calculate the determinants of the following matrices:

a) $A = \begin{bmatrix} -2 & 3 & 1 & 5 \\ 0 & 0 & 1 & 6 \\ 0 & 0 & 1 & 7 \\ 0 & 0 & 1 & 10 \end{bmatrix}$

b) $B = \begin{bmatrix} 3 & 2 & 5 & 4 \\ 1 & 0 & 6 & 1 \\ 1 & 0 & 7 & 1 \\ 1 & 0 & 10 & 1 \end{bmatrix}$

Solution

(a) Expand along 1st column of A:

$$\det(A) = -2(-1)^{1+1} \begin{vmatrix} 0 & 1 & 6 \\ 0 & 1 & 7 \\ 0 & 1 & 10 \end{vmatrix} = 0$$

because it has a zero column.

(a) Expand along 2nd column of B,

$$\det(B) = 2(-1)^{2+2} \begin{vmatrix} 1 & 6 & 1 \\ 1 & 7 & 1 \\ 1 & 10 & 1 \end{vmatrix} = 0$$

because it has 2 equal columns.

Question 8. 1. Transform the matrix $M = \begin{bmatrix} 1 & -1 & 4 \\ -1 & 3 & 2 \\ 3 & 1 & 1 \end{bmatrix}$ to row echelon form, then deduce the value of $\det(M)$.

2. Calculate the determinants of the following matrices:

(a) $A = \begin{bmatrix} 1 & -1 & 10 & 4 \\ -1 & 3 & 11 & 2 \\ 0 & 0 & -2 & 0 \\ 3 & 1 & 13 & 1 \end{bmatrix}$

(b) $B = \begin{bmatrix} 1 & -1 & 10 & -8 \\ -1 & 3 & 11 & -4 \\ 0 & 0 & -2 & 0 \\ 3 & 1 & 13 & -2 \end{bmatrix}$

Solution

1.

$$M = \begin{bmatrix} 1 & -1 & 4 \\ -1 & 3 & 2 \\ 3 & 1 & 1 \end{bmatrix}$$

$$\xrightarrow[R_3 \to R_3 - 3R_1]{R_2 \to R_2 + R_1} M_1 = \begin{bmatrix} 1 & -1 & 4 \\ 0 & 2 & 6 \\ 0 & 0 & 5 \end{bmatrix}$$

$$\det(M) = \det(M_1) = 1 \times 2 \times 5 = 10.$$

2.

(a) Expand along the 3rd row of A :

$$\det(A) = 0 + 0 + (-2)(-1)^{3+3} \begin{vmatrix} 1 & -1 & 4 \\ -1 & 3 & 2 \\ 3 & 1 & 0 \end{vmatrix} + 0$$

$$= -2\det(M) = -20.$$

(b) $A \xrightarrow{C_4 \to -2C_4} B$ so $\det(B) = -2\det(A) = 40$.

Question 9. Consider the matrix

$$U = \begin{bmatrix} 1 & -1 & -2 & 2 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 2 & 1 \\ 0 & -1 & 1 & -2 \end{bmatrix}$$

1. Calculate $\det(U)$
2. Calculate $UU^\top$.
3. Calculate the determinant of the following matrix

$$A = \begin{bmatrix} 10 & -2 & -2 & -5 \\ -2 & 1 & -1 & 2 \\ -2 & -1 & 5 & 0 \\ -5 & 2 & 0 & 6 \end{bmatrix}$$

Solution

1.

- $U \xrightarrow{R_2 \leftrightarrow R_4} V = \begin{bmatrix} 1 & -1 & -2 & 2 \\ 0 & -1 & 1 & -2 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & -1 \end{bmatrix}$

So $\det(U) = -\det(V)$.

- V is upper triangular, so $\det(V) =$

$$\begin{aligned} & 1 \times (-1) \times 2 \times (-1) \\ & = 2 \end{aligned}$$

Thus, $\det(U) = -2$.

2.

$$\begin{aligned} UU^\top &= \begin{bmatrix} 1 & -1 & -2 & 2 \\ 0 & 0 & 0 & -1 \\ 0 & 0 & 2 & 1 \\ 0 & -1 & 1 & -2 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 0 \\ -1 & 0 & 0 & -1 \\ -2 & 0 & 2 & 1 \\ 2 & -1 & 1 & -2 \end{bmatrix} \\ &= \begin{bmatrix} 10 & -2 & -2 & -5 \\ -2 & 1 & -1 & 2 \\ -2 & -1 & 5 & 0 \\ -5 & 2 & 0 & 6 \end{bmatrix} \end{aligned}$$

3 .

$$A = UU^\top. \text{ So } \det(A) = \det(U)\det\left(U^\top\right)$$
$$= \det(U)\det(U) = (-2) \times (-2) = 4.$$

Question 10. Calculate the adjoint matrix of

$$A = \begin{bmatrix} 0 & 1 & -2 \\ 0 & 2 & 0 \\ 1 & -1 & -2 \end{bmatrix}$$

then deduce that A is invertible and A^{-1}.

Solution

$$A_{11} = (-1)^{1+1}\begin{vmatrix} 2 & 0 \\ -1 & -2 \end{vmatrix} = -4$$

$$A_{12} = (-1)^{1+2}\begin{vmatrix} 0 & 0 \\ 1 & -2 \end{vmatrix} = 0$$

$$A_{13} = (-1)^{1+3}\begin{vmatrix} 0 & 2 \\ 1 & -1 \end{vmatrix} = -2$$

$$A_{21} = (-1)^{2+1}\begin{vmatrix} 1 & -2 \\ -1 & -2 \end{vmatrix} = 4$$

$$A_{22} = (-1)^{2+2}\begin{vmatrix} 0 & -2 \\ 1 & -2 \end{vmatrix} = 2$$

$$A_{23} = (-1)^{2+3}\begin{vmatrix} 0 & 1 \\ 1 & -1 \end{vmatrix} = 1$$

$$A_{31} = (-1)^{3+1}\begin{vmatrix} 1 & -2 \\ 2 & 0 \end{vmatrix} = 4$$

$$A_{32} = (-1)^{3+2}\begin{vmatrix} 0 & -2 \\ 0 & 0 \end{vmatrix} = 0$$

$$A_{33} = (-1)^{3+3}\begin{vmatrix} 0 & 1 \\ 0 & 2 \end{vmatrix} = 0$$

- $\text{adj}(A) = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{bmatrix}^\top$

$= \begin{bmatrix} -4 & 4 & 4 \\ 0 & 2 & 0 \\ -2 & 1 & 0 \end{bmatrix}$

- $\det(A) = 0A_{11} + 0A_{21} + 1A_{31}$

$= A_{31} = 4 \neq 0$

- $A^{-1} = \dfrac{1}{\det(A)} \text{adj}(A) = \begin{bmatrix} -1 & 1 & 1 \\ 0 & 1/2 & 0 \\ -1/2 & 1/4 & 0 \end{bmatrix}$